Intermediate Elective 2

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1. Inspiration

Visualization inspiration

I chose Steven Ponce's "Coal falls. Renewables rise." as my inspiration. I liked how the visualization uses two directly labeled time series lines, a dark background, and contrasting colors to show two categories changing through time. I also liked that the color contrast helps separate a less desirable trend from a more desirable one, which fits my goal of comparing non-native and native vegetation cover in vernal pools.

Why it makes sense for my dataset

This inspiration makes sense for my dataset because the vernal pool vegetation data include percent cover values across multiple years. Since I am interested in comparing native and non-native vegetation cover through time, a two-line time series is a good way to show whether these groups follow similar or contrasting patterns.

Variables and visual mapping

The x-axis will show year, and the y-axis will show mean percent cover across vernal pools. The `cover_group` variable will be mapped to color so native and non-native vegetation are shown as separate lines. Points will show yearly values, lines will show the trend through time, and direct labels/annotations will help emphasize the contrast between native and non-native cover.

2. Planned figure

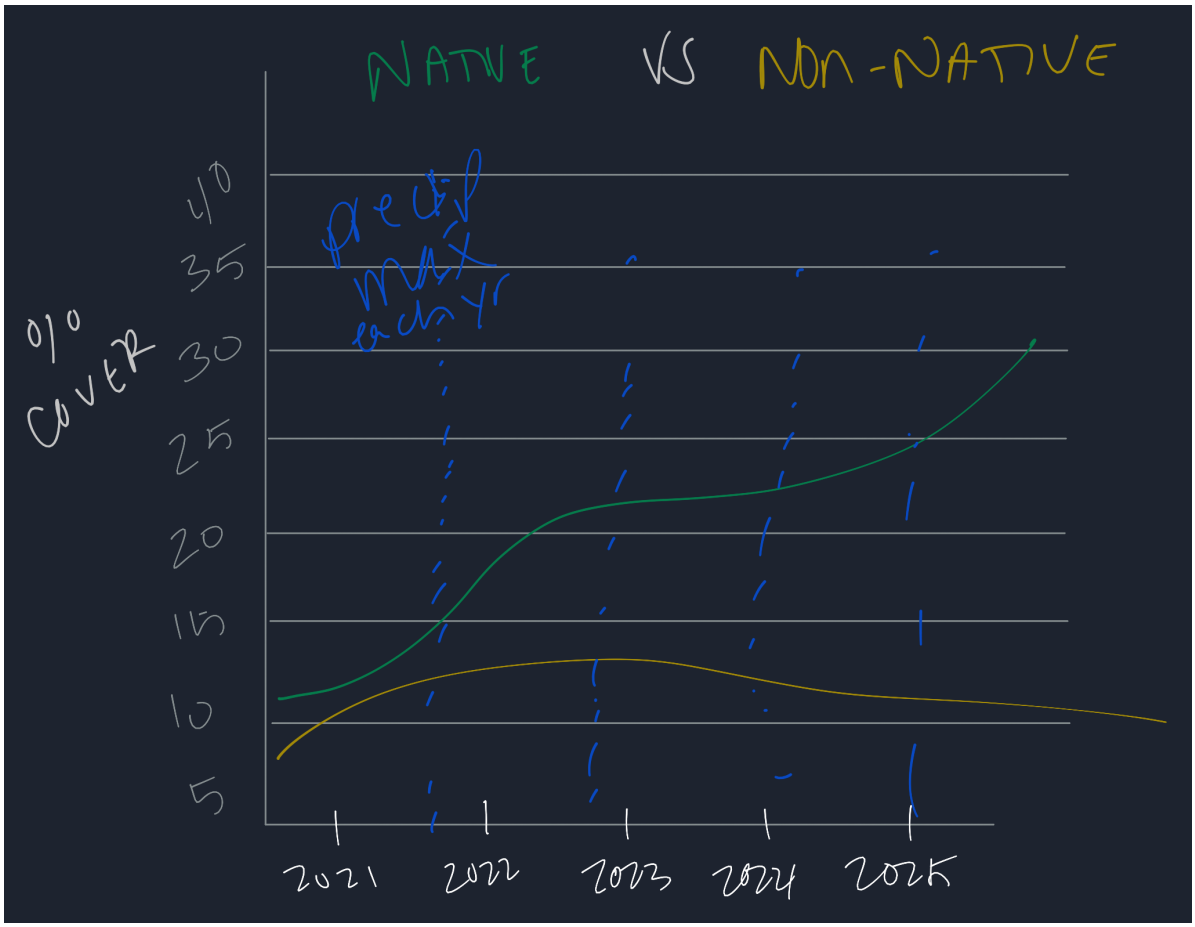


Figure 1: The sketch is a planned figure that includes two time series lines, direct labels for native and non-native cover, and a dark background inspired by the original visualization, Steven Ponce's "Coal falls. Renewables rise."

3. Code the figure

Setup

Packages

```
# general use
```

```r
library(tidyverse) # data cleaning, wrangling, and plotting
library(janitor) # clean column names
library(glue) # combine text and values for labels
library(scales) # format axis labels
library(lubridate) # clean dates

# file organization
library(here) # create reproducible file paths

# data visualization
library(naniar) # visualize missing data
library(patchwork) # arrange plots
library(viridis) # color palette
library(ggtext) # styled text in ggplot titles/subtitles
```

Data

```r
# read in vegetation data
veg <- read_csv(here("data", "veg.csv"))

# read in vernal pool vegetation metadata
metadata <- read_csv(here("data", "vp_veg_metadata.csv"))

# read in NOAA weather data for possible precipitation annotations
weather <- read_csv(here("data", "NOAA-weather-data.csv"))
```

Cleaning

Clean vegetation data

```r
# create cleaned vegetation data frame by cover type
cover_clean <- veg |>

  # keep only vernal pool transects at NCOS
  filter(str_detect(transect_name, "VP") &
           site == "ncos") |>

  # keep only native and non-native cover rows
```

```r
  filter(cover_category %in% c("NATIVE COVER", "NON-NATIVE COVER")) |>

  # keep only columns needed
  select(date,
         year,
         transect_name,
         cover_category,
         percent_cover) |>

  # remove rows with missing percent cover
  filter(!is.na(percent_cover)) |>

  # create month column
  mutate(month = month(date),

         # create simpler labels for plotting
         cover_group = case_when(
           cover_category == "NATIVE COVER" ~ "Native",
           cover_category == "NON-NATIVE COVER" ~ "Non-native"
         ))

# check cleaned data
cover_clean |>
  slice_sample(n = 10)
```

```
# A tibble: 10 x 7
   date        year transect_name cover_category percent_cover month cover_group
   <date>     <dbl> <chr>         <chr>                  <dbl> <dbl> <chr>
 1 2022-05-26  2022 VP-04         NON-NATIVE CO~             3     5 Non-
native
 2 2021-05-13  2021 VP-06         NON-NATIVE CO~             2     5 Non-
native
 3 2021-05-13  2021 VP-06         NATIVE COVER               1     5 Native
 4 2023-06-20  2023 VP-03         NATIVE COVER              10     6 Native
 5 2023-06-22  2023 VP-08         NON-NATIVE CO~             3     6 Non-
native
 6 2024-06-21  2024 VP-04         NATIVE COVER               1     6 Native
 7 2022-05-26  2022 VP-03         NATIVE COVER               1     5 Native
 8 2025-06-16  2025 VP-06         NON-NATIVE CO~            11     6 Non-
native
 9 2023-06-21  2023 VP-06         NON-NATIVE CO~             7     6 Non-
native
```

10 2022-05-31 2022 VP-07 NATIVE COVER 12 5 Native

Summarize cover by month and year

```
# summarize cover by year, month, pool, and cover group
cover_pool_month <- cover_clean |>

  # group by year, month, pool, and cover group
  group_by(year, month, transect_name, cover_group) |>

  # calculate mean percent cover for each pool/month/cover group
  summarize(mean_pc = mean(percent_cover, na.rm = TRUE),
            .groups = "drop")


# create month/year order for plotting
month_order <- cover_pool_month |>

  # keep unique year/month combinations
  distinct(year, month) |>

  # arrange in time order
  arrange(year, month) |>

  # create readable month/year label and month-date for plotting
  mutate(month_year = glue("{month.abb[month]} {year}"),
         month_date = make_date(year = year,
                                month = month,
                                day = 1))


# summarize average cover across pools by month and cover group
cover_month_summary <- cover_pool_month |>

  # group by year, month, and cover group
  group_by(year, month, cover_group) |>

  # calculate average cover across vernal pool transects
  summarize(mean_cover = mean(mean_pc, na.rm = TRUE),
            sd_cover = sd(mean_pc, na.rm = TRUE),
            n_pools = n_distinct(transect_name),
```

```r
            .groups = "drop") |>

  # join month/year info for plotting
  left_join(month_order, by = c("year", "month"))

# display summarized data
cover_month_summary
```

```
# A tibble: 16 x 8
    year month cover_group mean_cover sd_cover n_pools month_year month_date
   <dbl> <dbl> <chr>            <dbl>    <dbl>   <int> <glue>     <date>
 1  2021     5 Native            7.65     4.76       8 May 2021   2021-05-01
 2  2021     5 Non-native        3.41     1.08       8 May 2021   2021-05-01
 3  2022     5 Native            7.01     4.86       8 May 2022   2022-05-01
 4  2022     5 Non-native        4.12     1.78       8 May 2022   2022-05-01
 5  2022     6 Native           11.2     NA          1 Jun 2022   2022-06-01
 6  2022     6 Non-native        5.02    NA          1 Jun 2022   2022-06-01
 7  2023     6 Native           14.4      6.08       6 Jun 2023   2023-06-01
 8  2023     6 Non-native        9.77     3.11       6 Jun 2023   2023-06-01
 9  2023     7 Native           30.5     10.7        2 Jul 2023   2023-07-01
10  2023     7 Non-native        7.76     1.02       2 Jul 2023   2023-07-01
11  2024     6 Native           19.0      7.58       6 Jun 2024   2024-06-01
12  2024     6 Non-native       13.9      3.27       6 Jun 2024   2024-06-01
13  2024     7 Native           26.4     11.2        2 Jul 2024   2024-07-01
14  2024     7 Non-native       10.9      2.54       2 Jul 2024   2024-07-01
15  2025     6 Native           20.4     12.6        8 Jun 2025   2025-06-01
16  2025     6 Non-native        7.97     2.71       8 Jun 2025   2025-06-01
```

Clean weather data

```r
# clean NOAA precipitation data
weather_clean <- weather |>

  # clean column names
  clean_names() |>

  # change date column to standard date format
  mutate(date = mdy(date)) |>

  # keep precipitation column and create year/month columns
```

```r
  select(date, prcp) |>
  mutate(year = year(date),
         month = month(date))
```

Summarize precipitation by month

```r
# summarize total precipitation by year and month
monthly_precip <- weather_clean |>

  # keep years that overlap with vegetation data
  filter(year %in% unique(cover_month_summary$year)) |>

  # group by year, month, and plot date
  group_by(year, month) |>

  # calculate total monthly precipitation
  summarize(total_precip = sum(prcp, na.rm = TRUE),
            .groups = "drop")



# identify wettest month in each year within the plotted vegetation range
wettest_months_plot <- monthly_precip |>

  # create month-date column
  mutate(month_date = make_date(year = year,
                                month = month,
                                day = 1)) |>

  # keep months inside the plotted figure range
  filter(month_date >= make_date(2021, 5, 1),
         month_date <= make_date(2025, 6, 1)) |>

  # group by year
  group_by(year) |>

  # keep wettest month in each year
  slice_max(total_precip, n = 1, with_ties = FALSE) |>

  # ungroup data frame
```

```
  ungroup() |>

  # create label text for annotation
  mutate(label = glue("Wettest month\n{month.abb[month]} {year}:
   ↪  {round(total_precip, 1)} in"))
```

Final figure

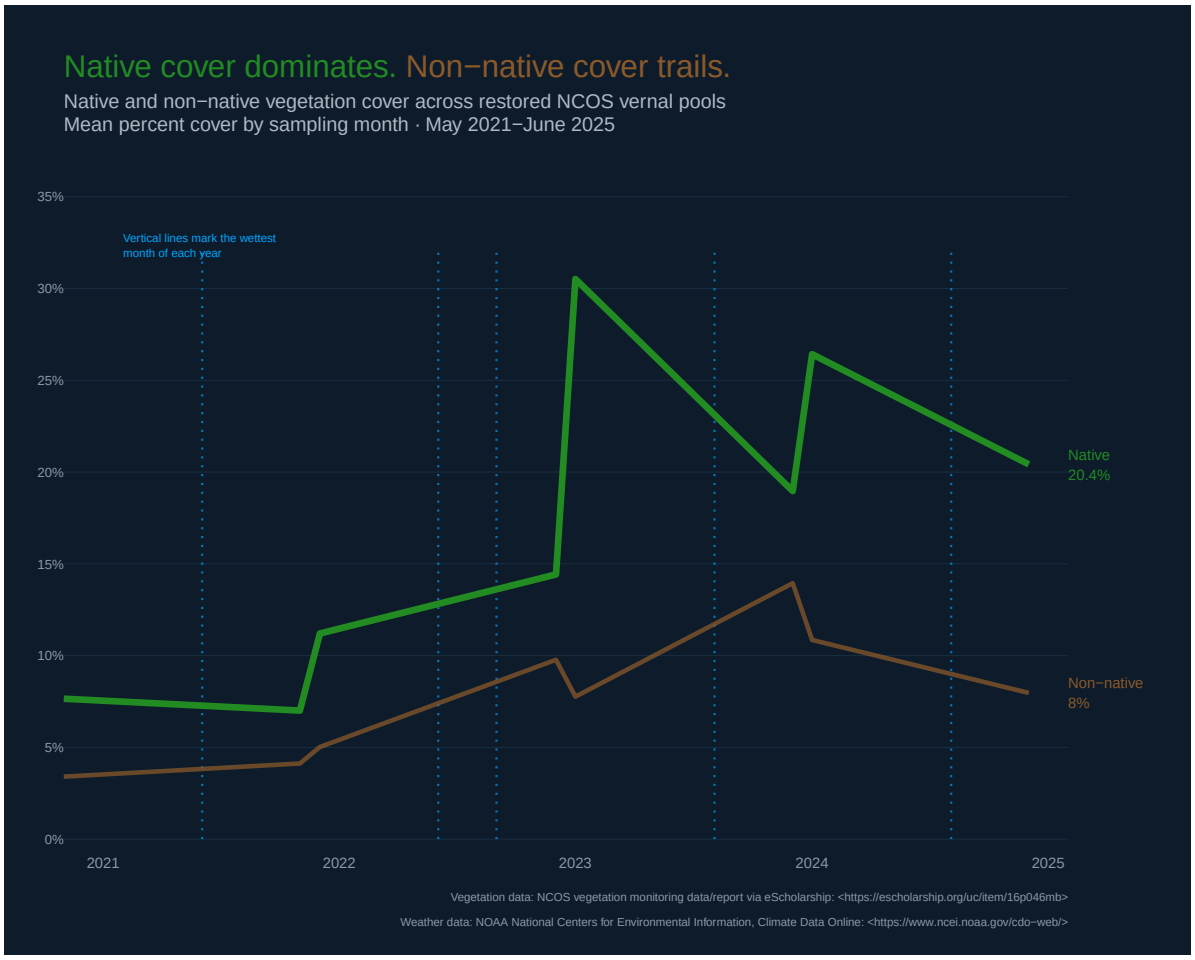
```
# create month-date columns for plotting
cover_month_summary <- cover_month_summary |>
  mutate(month_date = make_date(year = year,
                                month = month,
                                day = 1))


# identify wettest month in each year within the plotted vegetation range
wettest_months_plot <- monthly_precip |>

  # create month-date column
  mutate(month_date = make_date(year = year,
                                month = month,
                                day = 1)) |>

  # keep only months within the plot range
  filter(month_date >= make_date(2021, 5, 1),
         month_date <= make_date(2025, 6, 1)) |>

  # group by year
  group_by(year) |>

  # keep the wettest month in each year
  slice_max(total_precip, n = 1, with_ties = FALSE) |>

  # ungroup data frame
  ungroup()


# create endpoint labels for each line
end_labels <- cover_month_summary |>
```

```r
  # keep the last month-year for each cover group
  group_by(cover_group) |>
  filter(month_date == max(month_date)) |>

  # ungroup data frame
  ungroup() |>

  # create label and x position
  mutate(label = glue("{cover_group}\n{round(mean_cover, 1)}%"),
         label_date = month_date + months(2))


# adjusted variable colors to be more intuitive for vegetation data
bg_color <- "#0D1B2A"
native_green <- "forestgreen"
nonnative_brown <- "tan4"
text_white <- "#FFFFFF"
text_muted <- "#8A98A8"
text_sub <- "#B0B8C4"
grid_color <- "#1A2E42"
precip_line <- "#00A6FF"


# create styled title with colored text
title_text <- glue(
  "<span style='font-weight:900; color:{native_green};'>Native cover
   ↪  dominates. </span>",
  "<span style='font-weight:900; color:{nonnative_brown};'> Non-native cover
   ↪  trails.</span>"
)


# create styled subtitle
subtitle_text <- glue(
  "<span style='font-size:12pt; color:{text_sub};'>",
  "Native and non-native vegetation cover across restored NCOS vernal
   ↪  pools<br>",
  "Mean percent cover by sampling month · May 2021–June 2025",
  "</span>"
)


# create final plot inspired by Steven Ponce's visualization
```

```r
final_cover_plot <- ggplot(data = cover_month_summary,
                           mapping = aes(x = month_date,
                                         y = mean_cover,
                                         color = cover_group,
                                         group = cover_group)) +

  # add horizontal grid lines
  geom_hline(yintercept = seq(0, 35, by = 5),
             color = grid_color,
             linewidth = 0.25) +

  # add dotted vertical lines for the wettest precipitation month in each
   ↪ year
  geom_segment(data = wettest_months_plot,
               aes(x = month_date,
                   xend = month_date,
                   y = 0,
                   yend = 32),
               inherit.aes = FALSE,
               color = precip_line,
               linewidth = 0.63,
               linetype = "dotted",
               alpha = 0.65) +

  # add annotation for precipitation lines
  annotate("text",
           x = make_date(2021, 8, 1),
           y = 33,
           label = "Vertical lines mark the wettest\nmonth of each year",
           hjust = 0,
           vjust = 1,
           size = 2.5,
           color = precip_line,
           lineheight = 1.1) +

  # add native line with stronger emphasis
  geom_line(data = cover_month_summary |>
              filter(cover_group == "Native"),
            linewidth = 1.9,
            alpha = 1) +

  # add non-native line with softer emphasis
  geom_line(data = cover_month_summary |>
```

```
              filter(cover_group == "Non-native"),
            linewidth = 1.3,
            alpha = 0.75) +

  # add direct labels at the end of each line
  geom_text(data = end_labels,
            aes(x = label_date,
                label = label),
            hjust = 0,
            vjust = 0.5,
            size = 3.2,
            lineheight = 1.1,
            show.legend = FALSE) +

  # set native and non-native colors
  scale_color_manual(values = c("Native" = native_green,
                                "Non-native" = nonnative_brown)) +

  # set x-axis as actual month-date scale
  scale_x_date(date_labels = "%Y",
             breaks = make_date(2021:2025, 7, 1),
             limits = c(make_date(2021, 5, 1),
                        make_date(2025, 8, 1)),
             expand = expansion(mult = c(0, 0))) +

  # format y-axis as percent cover
  scale_y_continuous(limits = c(0, 35),
                     breaks = seq(0, 35, by = 5),
                     labels = \(x) glue("{x}%"),
                     expand = expansion(mult = c(0.01, 0.04))) +

  # add title, subtitle, and caption
  labs(title = title_text,
       subtitle = subtitle_text,
       caption = "Vegetation data: NCOS vegetation monitoring data/report via
        ↪  eScholarship: <https://escholarship.org/uc/item/16p046mb>

Weather data: NOAA National Centers for Environmental Information, Climate
 ↪  Data Online: <https://www.ncei.noaa.gov/cdo-web/>",
       x = NULL,
       y = NULL) +

  # use a dark theme similar to the inspiration figure
```

```r
  theme_void() +

  # customize theme elements
  theme(
    plot.background = element_rect(fill = bg_color, color = NA),
    panel.background = element_rect(fill = bg_color, color = NA),

    plot.title = element_markdown(color = text_white,
                                  size = 19,
                                  hjust = 0,
                                  lineheight = 1.1,
                                  margin = margin(t = 10, b = 6)),

    plot.subtitle = element_markdown(color = text_sub,
                                     size = 10,
                                     hjust = 0,
                                     lineheight = 1.4,
                                     margin = margin(b = 22)),

    plot.caption = element_text(color = text_muted,
                                size = 7,
                                hjust = 1,
                                margin = margin(t = 14)),

    axis.text.x = element_text(size = 9,
                               color = text_muted,
                               margin = margin(t = 6)),

    axis.text.y = element_text(size = 8,
                               color = text_muted,
                               hjust = 1),

    axis.ticks = element_blank(),
    panel.grid = element_blank(),
    legend.position = "none",
    plot.margin = margin(t = 20, r = 75, b = 16, l = 20)
  ) +

  # allow labels to extend beyond plot area
  coord_cartesian(clip = "off")

# display final plot
final_cover_plot
```




```r
# save final figure as a png file
ggsave(filename = here("images", "native_nonnative_cover.png"),
       plot = final_cover_plot,
       width = 8,
       height = 6,
       units = "in",
       dpi = 300)
```

4. Describe the process

Using someone else's code

I copied Steven Ponce's code and used it as a template for my own visualization. I kept some of the main design choices, like the dark background, direct labels, contrasting line colors, and annotation lines, but changed the data, variables, colors, and labels to fit my project.

Challenges

The biggest challenge was figuring out the right way to clean and summarize my data. I first tried going in the direction of explicit zeros and then an implicit/observed-species approach, but I realized that was more complicated than I needed because I was not comparing individual species. Since I was focused on measured native and non-native cover, it made more sense to use the existing cover_category column.

How this differs from previous figures

This figure required more setup for visual style before making the plot. I created separate objects for colors, title text, subtitle text, caption text, and annotation colors so the final figure would look more like the inspiration. I also used ggtext for styled title text and added precipitation annotation lines, which made the plot more customized than the figures I made previously.